EEG-Based Emotion Recognition for Hearing Impaired and Normal Individuals With Residual Feature Pyramids Network Based on Time–Frequency–Spatial Features

Fazheng Hou (侯法政), Junjie Liu (刘俊杰), Zhongli Bai (白忠立), Zhiyi Yang (杨知毅), Jiayin Liu (刘家寅), Qiang Gao (高强), and Yu Song (宋雨), *Member, IEEE*

***Abstract*— With the development of affective computing, discriminative feature selection is critical for electroencephalography (EEG) emotion recognition. In this article, we fused four EEG feature matrices constructed by the preprocessed signal, the differential entropy (DE), the symmetric difference, and the symmetric quotient based on the International 10–20 system, which integrates time-, frequency-, and spatial-domain information of EEG signals. For the feature classification model, we used the space-to-depth (S2D) layer instead of the convolutional neural network (CNN) as the backbone to reduce the calculation of the model without affecting the classification performance. The residual feature pyramid network (RFPN) was proposed to obtain the correlation of channels, and then, the deep multiscale semantic information of EEG feature maps is captured. The emotion classification strategy was evaluated by DEAP, SEED, SEED-IV, and our hearing-impaired EEG dataset (HIED). The classification accuracies were 93.56% (four-class, DEAP), 96.84% (three-class, SEED), 91.62% (four-class, SEED-IV), and 87.74% (six-class, HIED). Furthermore, we also found that the difference in emotional response between the left and right brain regions of hearing-impaired subjects is more obvious than that of normal subjects.**

***Index Terms*— Electroencephalography (EEG), emotion recognition, hearing-impaired subjects, residual feature pyramid network (RFPN), space-to-depth (S2D) layer.**

I. Introduction

WITH the development of computer technology and sensor technology, human–computer interaction (HCI) has made great progress [1]. Emotion recognition, as an important part of HCI, has become a research hotspot. The emotion can be recognized by nonphysiological signals (body postures [2], speech [3], or facial expression [4]) and physiological signals (electroencephalography (EEG) [5], EOG [6], and ECG [7]). Recent studies have shown that physiological signals can effectively reflect human real feelings [8]. Among them, the EEG signals have a higher temporal resolution and can reflect the activities of brain regions for different emotional states [9].

For EEG emotion recognition, the current methods were mainly focused on feature extraction and the construction of classification model. EEG features from the time domain, frequency domain, and time–frequency domain were widely used [10]. Among them, frequency-domain features, such as differential entropy (DE), power spectral density (PSD), and sample entropy (SE), have been proven their effectiveness in emotion recognition [11]. Liu et al. [12] proposed a dynamic DE (DDE) feature extraction algorithm, which combines the characteristics of empirical mode decomposition (EMD) and DE. The algorithm was applied to the SEED dataset for validation with an accuracy of 97.56%. An et al. [13] used DE features to construct 3-D EEG features and combined convolutional autoencoders (CAE) for emotion recognition. Fang et al. [14] fused DE and PSD features and used a multifeature deep forest (MFDF) model to classify human emotion. Zheng et al. [15] combined event-related potential (ERP) with improved multiscale SE (MMSE) as an EEG feature and then used a random forest classification algorithm to identify human emotions.

The time-, frequency-, or time–frequency-domain features only consider the variation of EEG amplitude with time and the distribution of energy in specific frequency bands. The researchers found that the different brain regions were associated with different emotions. Li et al. [16] used a spatial frequency convolutional self-attention network (SFCSAN) to capture and fuse spatial information and frequency-domain information for emotion recognition. Huang et al. [17] constructed EEG matrices to obtain the left and right brain region differences. However, the spatial domain features ignore the changes in EEG signals over time. Inspired by the previous research, we fused four EEG feature matrices constructed by the preprocessed signal, the DE, the symmetric difference, and

Manuscript received 8 August 2022; revised 7 January 2023; accepted 15 January 2023. Date of publication 27 January 2023; date of current version 7 February 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62103299 and in part by the 2021 Tianjin Postgraduate Research and Innovation Project under Grant 2021YJSS092. The Associate Editor coordinating the review process was Dr. John Lataire. *(Fazheng Hou and Junjie Liu are co-first authors.) (Corresponding authors: Yu Song; Qiang Gao.)*

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Ethics Committee at Tianjin University of Technology.

Fazheng Hou, Junjie Liu, Zhongli Bai, Zhiyi Yang, Jiayin Liu, and Yu Song are with the Tianjin Key Laboratory for Control Theory and Applications in Complicated Systems, School of Electrical Engineering and Automation, Tianjin University of Technology, Tianjin 300384, China (e-mail: hfzbrightsome@hotmail.com; ljjtju@163.com; ZL.Bai@hotmail.com; yzybri-lliant@hotmail.com; ljysweet@yeah.net; jasonsongrain@hotmail.com).

Qiang Gao is with the Tianjin Key Laboratory for Control Theory and Applications in Complicated Systems, TUT Maritime College, Tianjin University of Technology, Tianjin 300384, China (e-mail: gaoqiang@tjut.edu.cn).

Digital Object Identifier 10.1109/TIM.2023.3240230

the symmetric quotient, to obtain the time-, frequency-, and spatial-domain information of EEG signals.

In terms of emotion recognition models, the previous research focused on traditional machine learning classifiers, such as k-nearest neighbor (KNN), support vector machine (SVM), and decision tree (DT) [18]. Nowadays, many researchers are devoted to the study of emotion recognition based on deep learning. Al-Nafjan et al. [19] used a deep neural network (DNN) to extract PSD and frontal lobe asymmetry features of EEG for emotion recognition on the public dataset DEAP. Pan and Zheng [20] utilized a novel sample generation method of generative adversarial networks (GANs) to address the shortage of EEG samples and the imbalance between samples to improve the performance of emotion recognition. Liu et al. [21] used a spatiotemporal feature extraction module and EEG channel attention weight learning module to form a 3-D convolution attention neural network (3-DCANN), which adopted the fusion of spatiotemporal features and dual attention weights as features for emotion recognition. Ramzan and Dawn [22] integrated convolutional neural network (CNN) and LSTM to classify emotions and extract emotion-representative features. However, the recent classification model became much heavier as the deeper network layers, and it brings a great computational cost to use for emotion recognition. Hence, we carried out an emotion recognition model based on the residual feature pyramid network (RFPN) and space-to-depth (S2D) layer in this article. The S2D was applied to reduce the calculation of the model without affecting the classification performance. The proposed RFPN fully obtained the correlation features of channels and integrated the feature information among different scales.

Hearing-impaired people have difficulties in affective interaction to some extent, which may cause psychological or personality effects. In addition, there is relatively little research on emotion recognition based on hearing-impaired people. Therefore, in the previous work, we followed the experimental paradigm and the stimulus source of SEED [23] and MPED [24] to construct a hearing-impaired EEG dataset (HIED) based on three emotions (happiness, neutral, and sadness) [25], [26]. To further refine the multiple emotional states of the hearing impaired, we constructed an HIED based on six emotions (happiness, anger, encouragement, sadness, neutral, and fear). Four EEG feature matrices are constructed by the preprocessed signal, the DE, the symmetric difference, and the symmetric quotient, which integrates the time-, frequency-, and spatial-domain information of EEG signals. For the feature classification model, we used the S2D structure to replace the traditional convolutional network structure as the backbone network of the model. Also, based on Bi-FPN, a residual structure was added, which integrates the feature information between each scale and each node. Finally, we validated the model on the public datasets DEAP [27], SEED, SEED-IV [28], and HIED.

The rest content of this article is arranged as follows. The datasets and label rules are introduced in Section II. Section III describes the proposed method. The ablation experiments are carried out in Section IV. The experimental results are shown in Section V. We discussed and conclude the work in Section VI.

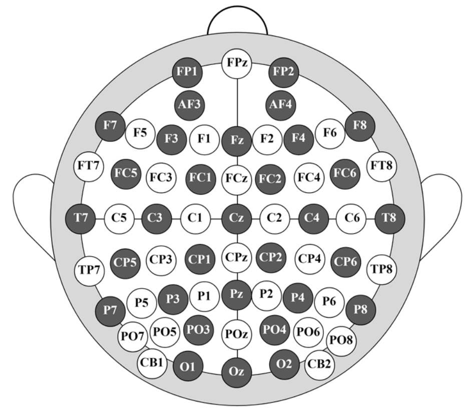

Fig. 1. International 10–20 system electrode positions, where the 32 dark electrodes are used in DEAP and all the 62 electrodes are used in SEED, SEED-IV, and HIED.

II. Experimental Setup

A. Datasets

In this article, we used DEAP, SEED, SEED-IV, and HIED databases to verify the proposed model. The DEAP dataset contains the EEG signals of 32 subjects when they are watching the 40 1-m music videos. Each trial was completed by the EEG acquisition device with 32 electrodes based on the International 10–20 system shown in Fig. 1. The subject was asked to give feedback by filling in the statistical table on different emotion dimensions (valence, arousal, dominance, and liking). The collected signals were downsampled from 512 to 128 Hz. For SEED dataset, it included the EEG signals of 15 subjects induced by 15 emotional movie clips (positive, neutral, and negative). Also, the SEED-IV dataset collected the EEG signals of 15 subjects induced by 24 emotional movie clips with four different emotional states (happiness, sadness, fear, and neutral). The corresponding EEG signals of SEED and SEED-IV were both obtained by 62 electrodes. Also, the average durations of the experiment are 240 s in SEED and 143 s in SEED-IV. The raw EEG signals were both downsampled to 200 Hz.

For the HIED dataset, we collected the EEG signals by SymAmps2 (Neuroscan, Australia) of 15 hearing-impaired subjects when they were watching six kinds of emotional movie clips (happiness, anger, encouragement, sadness, neutral, and fear). The subjects were recruited from the Technical College for the Deaf at Tianjin University of Technology. They were all hearing loss in both ears and needed to communicate with people through sign language or by wearing hearing aids.

In this experiment, we preselected ten clips for each emotion, and 15 postgraduates who majored in psychology were asked to score these clips. According to the scores of each clip, five video clips with the highest scores for each emotion were selected as the experimental stimulus materials, as shown in Table I. The average duration of the video stimulus was 231 s. The experimental process is shown in Fig. 2. There were 5 s for preparation before the video stimulus and 45 s for feedback after the stimulus. The feedback is embodied by subjects

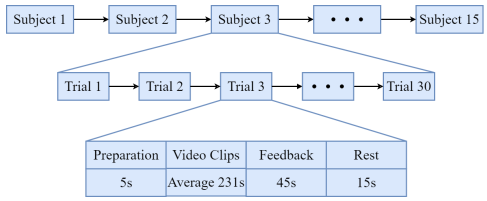

Fig. 2. Process of experiment.

TABLE I

SOURCE OF VIDEO STIMULI

No	Emotion	Video Clip Source	Duration (s)
1	E	Facing the Giants	373
2	S	Aftershock	205
3	A	Cry Me A Sad River	242
4	A	Shaolin Soccer	258
5	N	World Heritage in China	226
6	H	Lost In Thailand	238
7	E	Pioneer	269
8	E	Loke	198
9	S	Aftershock	205
10	A	Shaolin Soccer	210
11	A	The Matrix	284
12	N	World Heritage in China	221
13	H	Lost In Thailand	204
14	E	Leap	205
15	E	Pioneer	254
16	F	Deathly stillness	182
17	F	The Conjuring	200
18	A	Better Days	240
19	F	Coming Soon	192
20	S	1942	242
21	F	Coming Soon	189
22	H	Flirting Scholar	266
23	N	World Heritage in China	184
24	S	1942	240
25	F	Deathly stillness	211
26	N	World Heritage in China	240
27	H	Just Another Pandora's Box	241
28	S	1942	240
29	N	World Heritage in China	240
30	H	Just Another Pandora's Box	242

[1] H, A, E, S, N and F respectively represent happiness, anger, encouragement, sadness, neutral and fear.

being asked to complete scoring about their preferences and understanding, as well as evaluation of emotional reactions on the valence–arousal–dominance (V–A–D) dimension of the stimulus. A 15-s break between the adjacent trials was provided to allow the subjects to calm down. Ultimately, for each subject, we got 30 trials separately.

For EEG preprocessing, to facilitate the complexity of subsequent calculations, we downsampled the sampling frequency of EEG data from 1000 to 200 Hz. Therefore, in this experiment, we use the bandpass filter with 1–75 Hz to remove high- or low-frequency noise, and the notch filter with 50 Hz to remove industrial frequency interference. TP9 and TP10 were utilized as reference electrodes located on both sides of the mastoid in place of the in-line reference electrodes. Finally, we remove the artifacts by ICA, which is regarded as an effective method for separating artifacts.

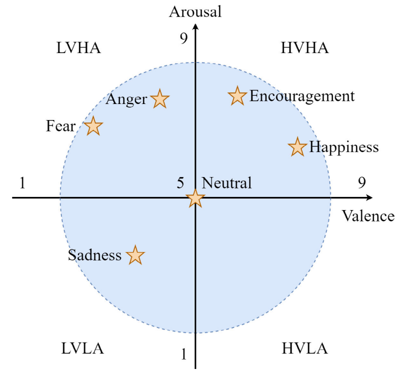

Fig. 3. Two-dimensional model of V/A.

B. Emotion Model and Label Rules

To facilitate model training and testing, we set labels according to the emotion model. In the current research, the V/A model is most commonly used [29]. The model can map discrete emotion states into a 2-D coordinate space, as shown in Fig. 3. The horizontal axis of the coordinate from left to right represents emotion from negative to positive. Also, the vertical axis of the coordinate from bottom to top represents the intensity of emotions from weak to strong. In previous research, emotions on a scale of 1–5 are considered low valence (LV) or low arousal (LA), and those of 6–9 are considered high valence (HV) or high arousal (HA).

In this work, we divided the data into HV/LV and HA/LA for the two-class experiments for the DEAP dataset. Moreover, four-classification experiments of DEAP were also carried out, such as HVHA, HVLA, LVHA, and LVLA. For SEED, according to the emotion labels provided by the dataset, the data were divided into three categories: positive, neutral, and negative. Furthermore, to ensure data consistency, we also removed neutral emotion segments from the dataset for positive and negative binary classification experiments. For SEED-IV, we divided the dataset into four categories by the provided labels from the description of SEED-IV: happiness, sadness, fear, and neutral. For HIED, we divided the data into six categories according to the corresponding emotional stimuli (happiness, anger, encouragement, sadness, neutral, and fear) for six-classification experiments. Moreover, the corresponding segments of three emotions (happiness, neutral, and sadness) were extracted for the subsequent three-classification experiments, and the label setting rules were the same as in the SEED dataset.

III. FEATURE EXTRACTION AND CLASSIFICATION

A. Feature Matrices Construction

In this step, four different feature matrices [preprocessed signal matrix (PSM), symmetric difference matrix (SDM), symmetric quotient matrix (SQM), and DE matrix (DEM)] were constructed based on the electrode positions provided by the International 10–20 system, as shown in Fig. 1, and the process of constructing different matrices is shown in Fig. 4.

1) Preprocessed Signal Matrix: According to the number of electrode channels, we construct multiple 1-D vectors $\boldsymbol{P}_l = [p_l^1, p_l^2, p_l^3, \cdots p_l^n]^T$ to represent the preprocessing EEG signals at the sample point l, where n is the number of channels,

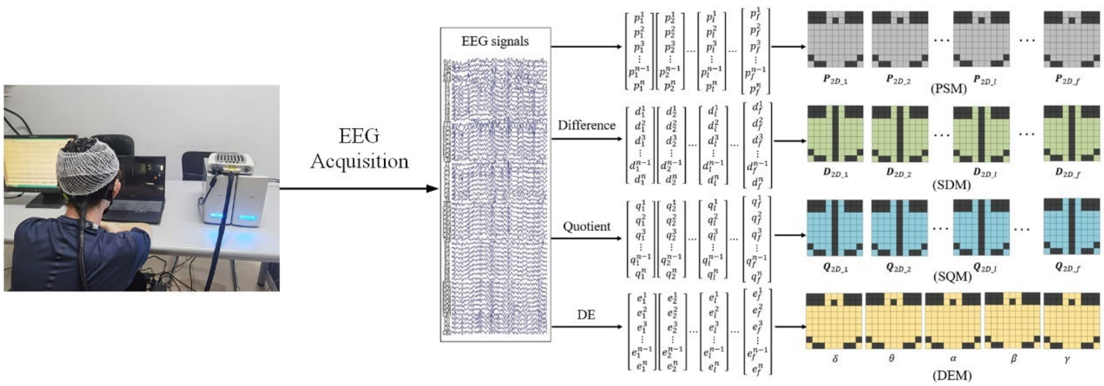

Fig. 4. Process of constructing different matrices, n represents the number of channels and f is the sample frequency.

$l \in [1, f]$, in which f is the sampling frequency. Also, n is 32 in DEAP, 62 in SEED, SEED-IV, and HIED. The number of 1-D vectors per experiment for each subject is $T \times f$ (T: length of the trial and f: the sampling frequency). However, 1-D vectors cannot represent the spatial information among electrodes. Therefore, to characterize the spatial information of the electrodes, we use the value of EEG to construct the 9×9 matrices named PSM based on the electrode positions. Besides, the PSM can also preserve the temporal information of EEG. The 2-D matrix $\boldsymbol{P}_{2D\_l}$ is given in (1), and the other positions in the matrix were set to 0 to avoid the interference of other factors.

We got $40 \times 60 \times 128$ preprocessed signal matrices for each subject by this method in DEAP (40 is the number of trials per subject, 60 is the duration of each trial, and 128 is the sampling frequency). In SEED, SEED-IV, and HIED, we obtained $15 \times 240/143/231 \times 200$ preprocessed signal matrices for each subject (15 trials, 240 or 143 or 231 s, 200 Hz)

$$\boldsymbol{P}_{2D\_l} = \begin{bmatrix} 0 & 0 & 0 & p_l^1 & p_l^2 & p_l^3 & 0 & 0 & 0 \\ 0 & 0 & 0 & p_l^4 & 0 & p_l^5 & 0 & 0 & 0 \\ p_l^6 & p_l^7 & p_l^8 & p_l^9 & p_l^{10} & p_l^{11} & p_l^{12} & p_l^{13} & p_l^{14} \\ p_l^{15} & p_l^{16} & p_l^{17} & p_l^{18} & p_l^{19} & p_l^{20} & p_l^{21} & p_l^{22} & p_l^{23} \\ p_l^{24} & p_l^{25} & p_l^{26} & p_l^{27} & p_l^{28} & p_l^{29} & p_l^{30} & p_l^{31} & p_l^{32} \\ p_l^{33} & p_l^{34} & p_l^{35} & p_l^{36} & p_l^{37} & p_l^{38} & p_l^{39} & p_l^{40} & p_l^{41} \\ p_l^{42} & p_l^{43} & p_l^{44} & p_l^{45} & p_l^{46} & p_l^{47} & p_l^{48} & p_l^{49} & p_l^{50} \\ 0 & p_l^{51} & p_l^{52} & p_l^{53} & p_l^{54} & p_l^{55} & p_l^{56} & p_l^{57} & 0 \\ p_l^{58} & 0 & 0 & p_l^{59} & p_l^{60} & p_l^{61} & 0 & 0 & p_l^{62} \end{bmatrix} \times \begin{pmatrix} \text{SEED} \\ \text{HIED} \end{pmatrix}$$

$$\text{or} = \begin{bmatrix} 0 & 0 & 0 & p_l^1 & 0 & p_l^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & p_l^3 & 0 & p_l^4 & 0 & 0 & 0 \\ p_l^5 & 0 & p_l^6 & 0 & p_l^7 & 0 & p_l^8 & 0 & p_l^9 \\ 0 & p_l^{10} & 0 & p_l^{11} & 0 & p_l^{12} & 0 & p_l^{13} & 0 \\ p_l^{14} & 0 & p_l^{15} & 0 & p_l^{16} & 0 & p_l^{17} & 0 & p_l^{18} \\ 0 & p_l^{19} & 0 & p_l^{20} & 0 & p_l^{21} & 0 & p_l^{22} & 0 \\ p_l^{23} & 0 & p_l^{24} & 0 & p_l^{25} & 0 & p_l^{26} & 0 & p_l^{27} \\ 0 & 0 & 0 & p_l^{28} & 0 & p_l^{29} & 0 & 0 & 0 \\ 0 & 0 & 0 & p_l^{30} & p_l^{31} & p_l^{32} & 0 & 0 & 0 \end{bmatrix} \times (\text{DEAP}). \tag{1}$$

In addition, in order to avoid the impact of the large numerical difference on the subsequent classification, we normalize each 2-D matrix by a normalization method called z-score, and the calculation formula is given as follows:

$$z_l^i = \frac{p_l^i - \bar{p}_l}{\sigma_l} \tag{2}$$

where $\bar{p}_l$ and σ_l represent the nonzero value mean and variance of the matrix corresponding to sampling point l, respectively.

2) Symmetric Difference Matrix: Neuroscience research shows that people's left and right brain regions respond differently to various emotions. Therefore, constructing the difference matrix of left and right brain regions is of great significance to improve the performance of emotion recognition. Based on the initial signal, we constructed the symmetric difference and SQMs based on sampling points.

First, remove the four channels (Fz, Cz, Pz, and Oz) for DEAP and eight channels (FPz, Fz, FCz, Cz, CPz, Pz, POz, and Oz) for SEED, SEED-IV, and HIED in the middle position from the preprocessed 1-D vector $\boldsymbol{P}_l = [p_l^1, p_l^2, p_l^3, \cdots p_l^n]^T$. For the convenience of subsequent calculation, the electrodes were renumbered. At this time, n is 28 for DEAP, and 54 for SEED, SEED-IV, and HIED. The symmetrical electrode pairs, such as F3–F4, CP1–CP2, and O1–O2, were found to get PSM. Also, their differences were calculated according to the following:

$$d_l^i = \begin{cases} p_l^i - p_l^{i+27}, & \text{if } i < 27 \\ p_l^i - p_l^{i-27}, & \text{if } i \geq 27 \end{cases} \begin{pmatrix} \text{SEED} \\ \text{HIED} \end{pmatrix}$$
$$\text{or} \begin{cases} p_l^i - p_l^{i+14}, & \text{if } i < 14 \\ p_l^i - p_l^{i-14}, & \text{if } i \geq 14 \end{cases} (\text{DEAP}). \tag{3}$$

Here, d_l^i represents the electrode pair difference value corresponding to the sampling point $l \in [1, f]$ and i represents the electrode serial number after removing the middle position channels. For example, d_1^1 is equal to channel FP1 (the number is 1) corresponding to the first sample point minus channel FP2 (the number is 15) in DEAP. In SEED, SEED-IV, and HIED, d_1^1 is equal to channel FP1 (the number is 1) corresponding to the first sample point minus channel FP2 (the number is 28). Hence, the 1-D vector $\boldsymbol{D}_l = [d_l^1, d_l^2, d_l^3, \cdots d_l^n]^T$ can be obtained to represent differences for symmetrical electrode pairs. Also, construct the SDM $\boldsymbol{D}_{2D\_l}$ named SDM in the same way as constructing the PSM, which is shown in the following formula:

$$
\begin{aligned}
\boldsymbol{D}_{2D\_l} = &\begin{bmatrix}
0 & 0 & 0 & d_l^1 & 0 & d_l^{28} & 0 & 0 & 0 \\
0 & 0 & 0 & d_l^2 & 0 & d_l^{29} & 0 & 0 & 0 \\
d_l^3 & d_l^4 & d_l^5 & d_l^6 & 0 & d_l^{33} & d_l^{32} & d_l^{31} & d_l^{30} \\
d_l^7 & d_l^8 & d_l^9 & d_l^{10} & 0 & d_l^{37} & d_l^{36} & d_l^{35} & d_l^{34} \\
d_l^{11} & d_l^{12} & d_l^{13} & d_l^{14} & 0 & d_l^{41} & d_l^{40} & d_l^{39} & d_l^{38} \\
d_l^{15} & d_l^{16} & d_l^{17} & d_l^{18} & 0 & d_l^{45} & d_l^{44} & d_l^{43} & d_l^{42} \\
d_l^{19} & d_l^{20} & d_l^{21} & d_l^{22} & 0 & d_l^{49} & d_l^{48} & d_l^{47} & d_l^{46} \\
0 & d_l^{23} & d_l^{24} & d_l^{25} & 0 & d_l^{52} & d_l^{51} & d_l^{50} & 0 \\
d_l^{26} & 0 & 0 & d_l^{27} & 0 & d_l^{54} & 0 & 0 & d_l^{53}
\end{bmatrix} \\
&\times \begin{pmatrix} \text{SEED} \\ \text{HIED} \end{pmatrix} \\
\text{or} = &\begin{bmatrix}
0 & 0 & 0 & d_l^1 & 0 & d_l^{15} & 0 & 0 & 0 \\
0 & 0 & 0 & d_l^2 & 0 & d_l^{16} & 0 & 0 & 0 \\
d_l^3 & 0 & d_l^4 & 0 & 0 & 0 & d_l^{18} & 0 & d_l^{17} \\
0 & d_l^5 & 0d_l^6 & 0 & d_l^{20} & 0 & d_l^{19} & 0 & \\
d_l^7 & 0 & d_l^8 & 0 & 0 & 0 & d_l^{22} & 0 & d_l^{21} \\
0 & d_l^9 & 0d_l^{10} & 0 & d_l^{24} & 0 & d_l^{23} & 0 & \\
d_l^{11} & 0 & d_l^{12} & 0 & 0 & 0 & d_l^{26} & 0 & d_l^{25} \\
0 & 0 & 0 & d_l^{13} & 0 & d_l^{27} & 0 & 0 & 0 \\
0 & 0 & 0 & d_l^{14} & 0 & d_l^{28} & 0 & 0 & 0
\end{bmatrix} \\
&\times (\text{DEAP}). \qquad (4)
\end{aligned}
$$

Similarly, in order to avoid the impact of too large numerical differences on the classification model, we still used the z-score as (3) for normalization to get the final SDM.

3) Symmetric Quotient Matrix: For the SQM construction method, the electrodes are also renumbered after removing the middle position channels. Analogously, q_l^i represents the quotient of the electrode pair corresponding to the sampling point l, and i represents the electrode serial number after removing the middle electrodes. For example, q_{10}^5 is equal to channel FC5 (the number is 5) corresponding to the tenth sampling point divided by channel FC6 (the number is 19) in DEAP. In SEED, SEED-IV, and HIED, q_{10}^5 is equal to channel FC5 (the number is 5) corresponding to the tenth sampling point divided by channel FC6 (the number is 32). From this, we got the 1-D vector $\boldsymbol{Q}_l = [q_l^1, q_l^2, q_l^3, \cdots q_l^n]^T$ of the symmetry quotient. We converted the 1-D vector $\boldsymbol{Q}_l$ to 2-D vector $\boldsymbol{Q}_{2D\_l}$ in the same way as PSM by following (4), where d_l^i is replaced by q_l^i. Of course, the matrices are normalized for improving the emotion recognition performance.

4) Differential Entropy Matrix: To ensure that the data volume is equal, we used 1-s time windows to extract the DE features of five frequency bands, which are delta (δ, $1-4$ Hz), theta (θ, $4-8$ Hz), alpha (α, $8-12$ Hz), beta (β, $12-30$ Hz), and gamma (γ, 3045 Hz). The formula for DE is given as follows:

$$
\text{DE} = -\int_a^b p(x)\log p(x)dx = \frac{1}{2}\log\left(2\pi e\sigma_i^2\right) \qquad (5)
$$

where $p(x)$ is a probability density function and follows a Gaussian distribution $N(\mu, \sigma_i^2)$ and $[a, b]$ represents the value range of a continuous signal over time.

We got DEMs, whose number is $T \times 5 \times n$ (T : length of each trial, 60 s for DEAP, 240 s for SEED, 143 s for SEED-IV, and 231 s for HIED; 5 : number of frequency bands; n : 32, 62, 62, and 62 for DEAP, SEED, SEED-IV, and HIED). The obtained DE features were also placed into a 9×9 matrix according to the electrode positions by following (1) and normalized by z-score.

B. Feature Fusion and Classification Model

For the emotion classification model, we applied the S2D layer to form the backbone network of the classification model and used the RFPN for cross-scale and cross-layer feature fusion, as shown in Fig. 5.

1) Feature Fusion Network: The feature fusion network was composed of the convolutional network. The four different feature matrices were fed into 1×1 convolution with 50 convolution kernels (not changing the size of feature matrices). After that, the batch normalization (BN) layer and the SiLU activation function were applied to the network to avoid possible gradient explosion problems and solve the problem of insufficient linear model capability. Therefore, we got feature maps of $50 \times 9 \times 9$ for each feature matrix. Furthermore, the nearest neighbor upsampling was applied to further capture the spatial information of EEG signals. Finally, the four upsampling matrices were attached to obtain a fusion feature matrix.

2) Space to Depth Layer: For the traditional FPN [30], most research considered the CNN as the backbone part of the FPN to obtain feature maps on different scales. However, the deep of backbone layers increases the computational cost of the model. Besides, FPN pays more attention to the fusion of high-level semantic information and low-level spatial information. Hence, inspired by the lightweight backbone of GiraffeDet [31], we considered the S2D structure (as shown in Fig. 6) to replace the traditional convolutional network as the backbone of the model. The S2D sampled and restructured feature maps at a fixed scale by transferring spatial dimension information to depth dimension. It replaces the traditional downsampling operation without additional parameters. We use the 1×1 convolution and SiLU activation function to output feature maps in fixed dimensions.

3) Residual Feature Pyramid Network: The FPN was proposed to integrate semantic information of different scales and complete the deep semantic feature extraction of multiscale information. Inspired by the FPN, Liu et al. [32] proposed the PANet, which added a layer of bottom-up paths on top of the FPN to solve the limitations of the unidirectional FPN. Tan et al. [33] proposed the BiFPN, adding a layer of cross-node fusion, which considered the information fusion between nodes. Hence, in this work, we applied a residual structure to construct an RFPN based on BiFPN. In RFPN, we used a residual structure to complete bidirectional cross-node fusion.

The RFPN includes two parts: residual connection and cross-scale fusion, as shown in Fig. 7. Residual connections can modularize pyramid units for simple stacking loops. Cross-scale fusion considered feature maps of the same scale and feature maps of adjacent scales, effectively avoiding large-scale changes in feature information. Taking the feature map M_5'' as an example, its final output includes the fusion of the nearest neighbor upsampling of the previous layer M_6', the previous

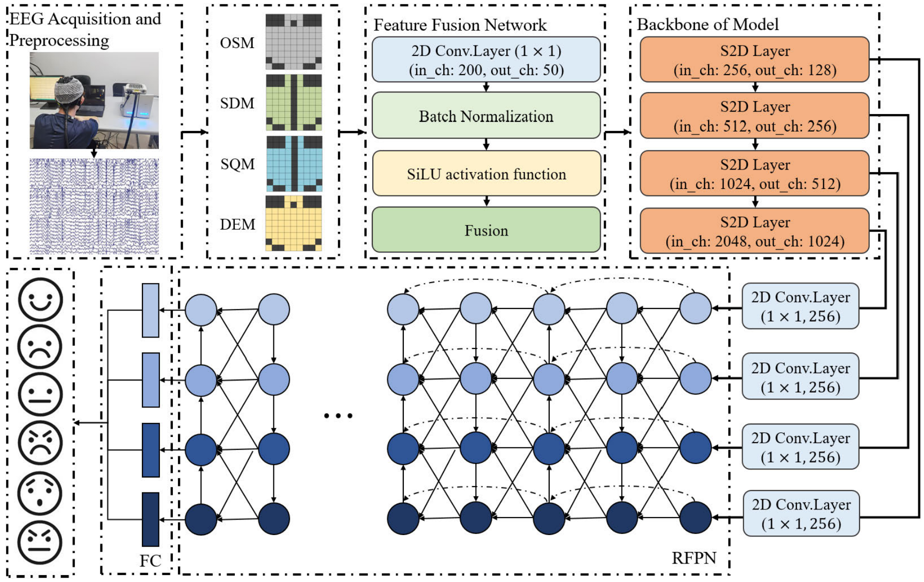

Fig. 5. Schematic of the overall framework.

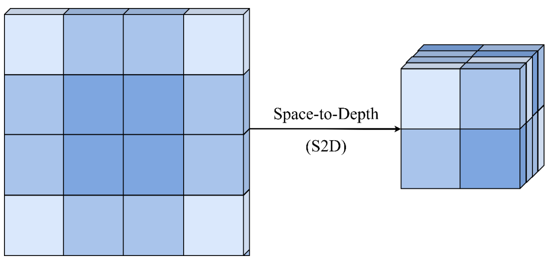

Fig. 6. S2D structure.

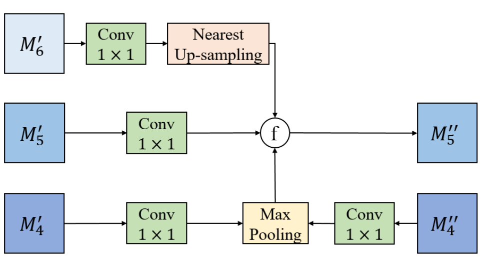

Fig. 7. Cross-scale fusion.

layer M_5', the downsampling by max pooling of the previous layer M_4', and the same layer M_4''. Here, we performed initial computation on feature matrices through the 3×3 convolutional network, BN, and SiLU activation function. The above operation was carried out twice, effectively using convolution to extract the spatial information of the matrices. After that, through four iterations of the S2D layer, four feature matrices of different scales M_3, M_4, M_5, and M_6 (the corresponding scales are, respectively, 15×15, 7×7, 3×3, and 1×1) were obtained. Then, after using the 1×1 convolution operation to unify the number of channels of the feature maps (the optimal number of channels is 256), put it into the RFPN to iterate, and fully obtain the fusion information between scales and between nodes. The final prediction result was obtained through the classification prediction network, which is the fully connected neural network (FC). The specific parameters of the network are shown in Table II.

TABLE II
HYPERPARAMETERS OF MODEL

Hyperparameter	Value
Batch size	32
Optimizer	SGD
Learning Rate	0.01
Step Decrease Ratio	0.01
Weight Decay	1×10^{-3}
Input Size	[9, 9]
S2D Down Scales	2
RFPN Input Channels	[128, 256, 512, 1024]
RFPN Output Channels	[256, 256, 256, 256]
Number of model loops	3
Training Epochs	20

TABLE III
TRAINING AND TEST SAMPLES IN EACH DATASET FOR EACH SUBJECT

Dataset	Experiment Type	Training Samples	Test Samples	Total Samples
DEAP	2-Class (HV/LV) 2-Class (HA/LA) 4-Class (HVHA, LVHA, HVLA, LVLA)	1680	720	2400
SEED	3-Class (positive/neutral/negative)	2375	1019	3394
	2-Class (positive/negative)	1556	668	2224
SEED-IV	4-class(H/F/N/S)[1]	2404	1031	3435
HIED	3-Class (positive/neutral/negative)	2403	1030	3433
	6-Class (H/A/E/S/N/F)[1]	4858	2082	6940

IV. SUBJECT-DEPENDENT EXPERIMENTAL RESULTS

In this work, different models were built for each subject and the final result is obtained by the mean results of 70–30 cross validation. The numbers of training and test samples in each dataset for each subject are shown in Table III. The

TABLE IV
STRUCTURE OF S2D CHAIN USED IN OUR EXPERIMENTS

Layer	Number of Filters	Size of Filters	Stride Value	Padding Value	Size of Output Features	
					DEAP	SEED/ HIED/ SEED-IV
Input	—	—	—	—	$9 \times 9 \times 128$	$9 \times 9 \times 200$
Up-sampling Layer	—	—	—	—	$64 \times 64 \times 128$	$64 \times 64 \times 200$
Conv Layer	32	3×3	2×2	1×1	$32 \times 32 \times 32$	$32 \times 32 \times 32$
Conv Layer	64	3×3	1×1	1×1	$32 \times 32 \times 64$	$32 \times 32 \times 64$
S2D	—	—	—	—	$16 \times 16 \times 256$	$16 \times 16 \times 256$
Conv Layer	128	1×1	1×1	0×0	$16 \times 16 \times 128$	$16 \times 16 \times 128$
S2D	—	—	—	—	$8 \times 8 \times 512$	$8 \times 8 \times 512$
Conv Layer	256	1×1	1×1	0×0	$8 \times 8 \times 256$	$8 \times 8 \times 256$
S2D	—	—	—	—	$4 \times 4 \times 1024$	$4 \times 4 \times 1024$
Conv Layer	512	1×1	1×1	0×0	$4 \times 4 \times 512$	$4 \times 4 \times 512$
S2D	—	—	—	—	$2 \times 2 \times 2048$	$2 \times 2 \times 2048$
Conv Layer	1024	1×1	1×1	0×0	$2 \times 2 \times 1024$	$2 \times 2 \times 1024$

model is built with the Pytorch framework and implemented in PyCharm software. The CPU of the computer is six cores and 12 threads, the computer's graphics card is NVIDIA GeForce RTX 3050 Ti Laptop GPU, and the computer's synchronous graphics DRAM is 4G. The model is run by GPU to save training time. Besides, the input scales of the model are shown in Table IV, and the structure of S2D used in our experiments and the size of output features in each layer are also added in detailed instructions.

A. Performance of Feature Fusion Strategy

To verify the feature fusion strategy, we constructed 14 experimental paradigms based on different combinations for ablation experiments on the three datasets. The experimental results are shown in Table V. We found that when we use a single matrix, the SDM performs best, which reached above 85% in the two-class classification experiment on DEAP and SEED datasets. In addition, the performance of the multiclassification task based on the SDM matrix is also better than other feature matrices on the used three datasets. On the DEAP and SEED, the fusion strategy of SDM and SQM achieved more than 90% in two- and three-class classification experiments, which outperformed other double matrix fusion strategies. The average accuracy of fusion based on SDM, SQM, and DEM can achieve a performance of around 95% in the two-class classification experiment on SEED, which even reached around 85% in the six-class classification experiments on HIED. It is worth noting that the strategy of fusing SDM, SQM, and DEM has reached an accuracy of 95.52% in the two-class classification experiment of SEED. Nevertheless, the fusion method of the four matrices proposed outperformed other fusion strategies. In DEAP, the two-class classification experiment based on subject-dependent achieved average accuracies of 96.89% in the valence, 96.82% in the arousal, and an average accuracy of 93.56% in the four-class experiment. The accuracy of the three- and two-class classification experiments reached 96.84% and 98.59% in the SEED dataset, respectively. For HIED, the average accuracy of the six- and three-class classification experiments reached 87.74% and 95.48%, respectively.

B. Comparison of Different Backbones

To demonstrate the advantage of the S2D layer, we used ResNet18 as the backbone of the network to build a classification model. Similarly, we unified the number of network layers to increase confidence in the experimental results. The results are shown in Table VI. Compared with ResNet18 as the backbone of the network, the S2D layer has a better performance, and the training time of the model is significantly reduced. In the SEED, SEED-IV, and HIED with more data samples, the training time of the model with S2D has been reduced more significantly, which is about 30 s shorter for each subject.

C. Evaluation of RFPN

To evaluate the proposed RFPN, comparative experiments were carried out on four datasets (DEAP, SEED, SEED-IV, and HIED). We built three classification models based on different FPN structures (FPN, PANet, and BiFPN). The results are shown in Table VII. We constructed the four networks in the same depth. It found that the proposed feature pyramid structure (RFPN) has improved performance by over 10% compared to the original FPN architecture on the used three datasets. For PANet and BiFPN, the performance improvement effect on EEG-based emotion recognition is not significant (about 2% of PANet and 8% of BiFPN higher than FPN) in the emotion classification experiments based on DEAP, SEED, SEED-IV, and HIED.

V. DISCUSSION

In this article, we constructed and fused four different matrices (PSM, SDM, SQM, and DEM) to obtain discriminative features for EEG-based emotion recognition. The PSM retained the information on the time domain of EEG. The SDM and SQM obtained spatial information and effectively captured the different responses of the left and right brain regions. Besides, in existing research, we found that DE is more effective than other frequency-domain features for EEG-based emotion recognition [34], [35]. Hence, we used the DE feature to build the fourth feature matrices to obtain the frequency-domain information of EEG. From Table V, we found that the SDM performs best when the input is a single matrix. Compared with the quotient value, the different values of the left and right brain regions can better represent

TABLE V

CLASSIFICATION RESULTS OF EXPERIMENTAL PARADIGMS BASED ON TENFOLD CROSS VALIDATION

Paradigm	Accuracy (%)						
	DEAP			SEED		HIED	
	2-Class (HV/LV)	2-Class (HA/LA)	4-Class (HVHA, LVHA, HVLA, LVLA)	3-Class (positive/ neutral/ negative)	2-Class (positive/ negative)	3-Class (positive/ neutral/ negative)	6-Class (H/A/E/S/N/F)
PSM	87.17	87.59	83.42	86.80	88.24	77.81	72.66
SDM	87.28	88.33	84.64	87.23	88.46	82.78	71.47
SQM	86.79	85.56	83.86	81.65	83.93	81.90	74.78
DEM	75.99	77.22	73.93	78.39	81.96	72.15	69.16
PSM+SDM	89.17	91.35	86.33	89.63	91.71	88.28	80.90
PSM+SQM	88.75	90.56	86.25	88.86	91.46	86.72	76.53
PSM+DEM	88.19	89.89	85.36	88.73	89.26	84.94	75.33
SDM+SQM	90.24	92.03	87.64	90.14	92.42	89.74	83.17
SDM+DEM	88.25	90.21	85.67	88.75	89.82	85.83	76.49
SQM+DEM	87.89	89.13	84.89	87.74	89.11	83.39	75.21
PSM+SDM+SQM	94.24	92.67	88.61	91.07	95.07	93.37	83.91
PSM+SDM+DEM	93.36	92.35	87.52	90.14	92.83	91.68	83.79
PSM+SQM+DEM	90.13	91.63	86.88	89.83	91.98	88.83	82.03
SDM+SQM+DEM	94.85	92.79	89.23	91.19	95.52	93.88	85.42
PSM+SDM+SQM+DEM	**96.89**	**96.82**	**93.56**	**96.84**	**98.59**	**95.48**	**87.74**

TABLE VI

ACCURACY AND TRAINING TIME OF RESNET AND S2D AS BACKBONE

Paradigm	Average Accuracy (%)							
	DEAP			SEED		SEED-IV	HIED	
	2-Class (HV/LV)	2-Class (HA/LA)	4-Class (HVHA, LVHA, HVLA, LVLA)	3-Class (positive/ neutral/ negative)	2-Class (positive/ negative)	4-class (H/F/N/S)	3-Class (positive/ neutral/ negative)	6-Class (H/A/E/S/N/F)
S2D+FPN	79.76	79.97	76.79	86.67	87.70	83.51	85.95	76.32
S2D+PANet	84.13	81.22	74.47	87.61	89.74	87.48	88.89	80.81
S2D+BiFPN	88.74	88.74	89.19	90.61	94.04	89.26	91.25	82.97
S2D+RFPN	**96.89**	**96.82**	**93.56**	**96.84**	**98.59**	**91.62**	**95.48**	**87.74**

the difference between the left and right brain regions in responding to different emotional states. In addition, the fusion of SDM and SQM is more effective for emotion recognition than the pairwise combination of other matrices, which may be because SQM and SDM have good complementarity and can mutually improve the differences in the responses of the left and right brain regions to different emotions. The satisfactory results achieved by the fusion of SDM, SQM, and DEM illustrate the importance of frequency-domain information for emotion recognition. Although various feature fusion strategies achieve higher accuracy, the proposed fusion strategy based on four feature matrices achieved the best classification performance.

We utilized the S2D as the backbone, and the training time was reduced for three datasets, as shown in Table VI. Besides, we obtained the number of parameters and the multiply–accumulate operations (MACs) of the model and compared them with some common networks in a similar parameter size. The results of comparison on the DEAP dataset for four-class experiment are shown in Table VIII. It can be seen that the number of model parameters and the computational complexity of the model are significantly reduced. The reason may be that S2D uses a fixed scale to transfer feature information from the spatial dimension to the channel dimension to complete the downsampling and reconstruction of the feature map. Compared to the complex operations of CNN with the domain-shift problems of the network as the backbone complexity increases, S2D is a simple dimensional reconstruction of the feature map and does not add additional network parameters such as weights ω and biases b that are carried in convolutional networks. In particular, for EEG-based emotion recognition, the S2D structure can better obtain correlation information of EEG channels than the convolutional network, which can effectively improve the classification performance.

It can be concluded that the proposed feature pyramid structure (RFPN) has a better performance than other structures in Table VII. The reason for achieving satisfactory results may be that RFPN fully integrates the feature semantic information among layers and nodes. On the one hand, the application of residual structure can perform information fusion between nodes of feature maps through shortcuts. On the other hand, the gradient explosion caused by the increase in the number of loops for the FPN was avoided, and the feature extraction

TABLE VII
CLASSIFICATION RESULTS OF PYRAMID STRUCTURES BASED ON TENFOLD CROSS VALIDATION

Datasets	Experiment Types	Backbone	Accuracy (%)	Training Time (s)
DEAP	2-Class (HV/LV)	ResNet	92.33	132.8
		S2D	96.89	110.3
	2-Class (HA/LA)	ResNet	93.54	133.2
		S2D	96.82	112.6
	4-Class (HVHA/HVLA LVHA/LVLA)	ResNet	86.47	135.7
		S2D	93.56	111.1
SEED	3-Class (positive/ neutral/ negative)	ResNet	94.63	188.6
		S2D	96.84	157.1
	2-Class (positive/ negative)	ResNet	95.38	187.3
		S2D	98.59	156.3
SEED-IV	4-class (H/F/N/S)[1]	ResNet	90.06	183.4
		S2D	91.62	155.3
HIED	3-Class (positive/ neutral/ negative)	ResNet	94.28	186.9
		S2D	95.48	159.6
	6-Class (H/A/E/S/N/F)	ResNet	87.74	187.7
		S2D	88.46	154.8

TABLE VIII
COMPARISON WITH STATE-OF-THE-ART CNNS AS BACKBONE ON DEAP DATASET FOR FOUR-CLASS EXPERIMENTS

Backbone	Params (M)	MACs (G)	Accuracy (%)
Resnet 18	17.97	1.88	86.47
Alexnet	67.38	0.78	82.56
Squeezenet 1_0	7.56	0.89	83.75
Squeezenet 1_1	7.52	0.42	84.58
CNN	9.46	0.32	89.27
S2D	**3.18**	**0.26**	**93.56**

capability of the network was increased. In addition, the performance of PANet and BiFPN was better than the original FPN, which proved the effectiveness of applying bidirectional structure and fusion features between nodes in FPN. Moreover, we compared the proposed model with current research, as shown in Table IX. It shows that the proposed method achieves more satisfactory results than others.

To verify the discrimination power of the proposed method, the features visualization maps are shown in Fig. 8 by t-SNE before and after training. It can be seen that after being trained by the proposed network, the distribution of emotion samples is more obvious, and only a few samples are disordered. It is further demonstrated that the proposed network can effectively extract discriminative features of EEG, which is important for emotion classification tasks.

To explore the difference between the left and right brain regions of normal subjects and hearing-impaired subjects in response to different emotional states, we compared the HIED dataset with the SEED dataset. We drew brain topographic maps of normal people and hearing-impaired people under the same emotional stimulation state, as shown in Fig. 9. We can see that, whether it is a positive, negative, or neutral emotional state, the difference between the left and right brain regions of the hearing impaired is more obvious, which has been proven by some studies [25], [46]. This may be why after the fusion of the difference matrices (SDM and SQM), the enhancing effect of the three-class experiment accuracy for hearing-impaired people (17.67% higher than PSM and 23.33% higher than DEM) is more obvious than that of hearing-normal people (10.04% higher than PSM and 18.45% higher than DEM).

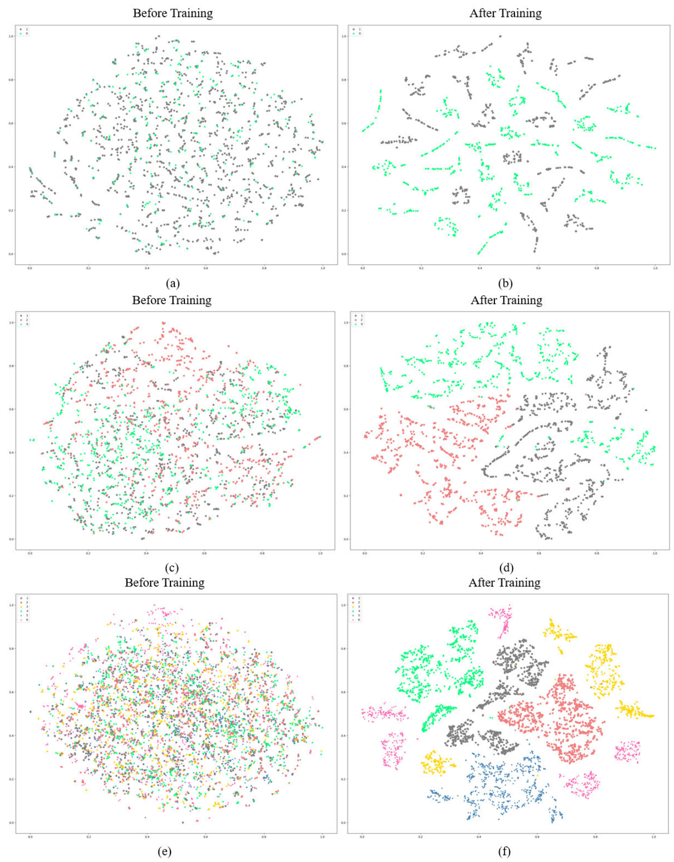

Fig. 8. Visualization maps by t-SNE on three datasets. (a) and (b) Visualization of DEAP for two-class experiment on valence. (c) and (d) Visualization of SEED for three-class experiment. (e) and (f) Visualization of HIED for six-class experiment.

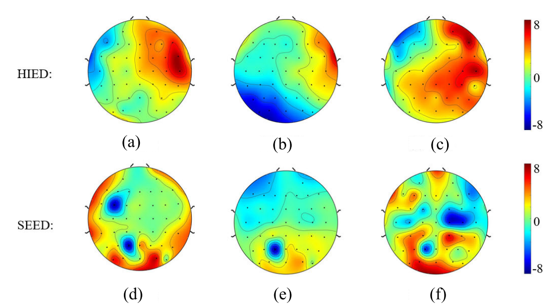

Fig. 9. Brain topographic maps of subject in HIED and SEED. (a)–(c) Three types of emotional brain maps of hearing-impaired people. (d)–(f) Brain maps of hearing-normal people.

TABLE IX
COMPARISON OF OTHER WORKS (%)

Datasets	Methods	Accuracy	F1 scores	Precision
DEAP	MM-mDistEn; ANN [36]	V: 90.26 A: 80.48	V: 86.03 A: 84.00	V: 85.34 A: 81.12
	Multi-Column CNN [37]	V: 90.01 A: 90.65	V: 82.79 A: 83.75	V: 85.57 A: 86.66
	DE; GCNN+LSTM [38]	V: 90.45 A: 90.60	V: 91.08 A: 90.94	—
	ST-GCLSTM [39]	V: 95.52 A: 95.04	V: 96.00 A: 95.00	—
	PSM, SDM, SQM, DEM; S2D+RFPN	V: **96.89** A: **96.82** 4: **93.56**	V: **96.06** A: **96.68** 4: **91.23**	V: **97.33** A: **95.42** 4: **93.26**
SEED	LMD [40]	3:94.87	3:95.00	—
	ATDD-LSTM [41]	3:90.92	—	—
	SCSTM, DSSWGFK [42]	3:82.29	3:82.00	—
	DE-CNN-BiLSTM [43]	3:94.82	—	—
	PSM, SDM, SQM, DEM; S2D+RFPN	3: **96.84** 2: **98.59**	3: **94.38** 2: **98.65**	3: **95.16** 2: **93.24**
SEED-IV	OGSSL [44]	4:81.29	—	—
	DDA [45]	4:80.82	—	—
	PSM, SDM, SQM, DEM; S2D+RFPN	4: **91.62**	4: **92.93**	4: **90.56**

VI. CONCLUSION

In this article, a novel feature fusion strategy integrating time–frequency–spatial-domain information and a classification model based on RFPN are proposed for emotion recognition. Four feature matrices (PSM, SDM, SQM, and DEM) were conducted and fused to represent different emotional state information. S2D was introduced to replace the traditional CNN as the backbone of the classification model to reduce model calculation cost and time. RFPN, composed of cross-layer connections based on residual structure and cross-scale fusion, was proposed to complete the deep semantic feature extraction of multiscale EEG feature maps. The proposed model was carried out on the DEAP, SEED, SEED-IV, and HIED datasets. The average accuracies were 93.56% (four-class on DEAP), 96.84% (three-class on SEED), 91.62% (four-class on SEED-IV), and 87.74% (four-class on HIED). In addition, through a comparison of HIED and SEED, we found that the difference between the left and right brain regions is more discriminative for the EEG-based emotion recognition of hearing-impaired people than normal people. In the future, we will select discriminative features for cross-subject experiments and online emotion recognition.

REFERENCES

[1] P. Sarma and S. Barma, "Emotion recognition by discriminating EEG segments with high affective content from automatically selected relevant channels," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–12, 2022.

[2] M. Gavrilescu, "Recognizing emotions from videos by studying facial expressions, body postures and hand gestures," in *Proc. 23rd Telecommun. Forum Telfor (TELFOR)*, Nov. 2015, pp. 720–723.

[3] Y. Ahn, S. J. Lee, and J. W. Shin, "Cross-corpus speech emotion recognition based on few-shot learning and domain adaptation," *IEEE Signal Process. Lett.*, vol. 28, pp. 1190–1194, 2021.

[4] K. Mohan, A. Seal, O. Krejcar, and A. Yazidi, "Facial expression recognition using local gravitational force descriptor-based deep convolution neural networks," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–12, 2021.

[5] Y. Peng, Y. Zhang, W. Kong, F. Nie, B.-L. Lu, and A. Cichocki, "S3LRR: A unified model for joint discriminative subspace identification and semisupervised EEG emotion recognition," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–13, 2022.

[6] B. H. Yilmaz and C. Kose, "A novel signal to image transformation and feature level fusion for multimodal emotion recognition," *Biomed. Eng.*, vol. 66, no. 4, pp. 353–362, Apr. 2021.

[7] A. Goshvarpour, A. Abbasi, and A. Goshvarpour, "An accurate emotion recognition system using ECG and GSR signals and matching pursuit method," *Biomed. J.*, vol. 40, no. 6, pp. 355–368, Dec. 2017.

[8] Z. Wang, Y. Wang, J. Zhang, C. Hu, Z. Yin, and Y. Song, "Spatial–temporal feature fusion neural network for EEG-based emotion recognition," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–12, 2022.

[9] Z. Gao, R. Li, C. Ma, L. Rui, and X. Sun, "Core-brain-network-based multilayer convolutional neural network for emotion recognition," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–9, 2021.

[10] X.-W. Wang, D. Nie, and B.-L. Lu, "EEG-based emotion recognition using frequency domain features and support vector machines," in *Proc. Int. Conf. Neural Inf. Process.*, in Lecture Notes in Computer Science: Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics, vol. 7062, 2011, pp. 734–743.

[11] W. Zheng, "Multichannel EEG-based emotion recognition via group sparse canonical correlation analysis," *IEEE Trans. Cogn. Develop. Syst.*, vol. 9, no. 3, pp. 281–290, Sep. 2017.

[12] S. Liu, X. Wang, L. Zhao, J. Zhao, Q. Xin, and S.-H. Wang, "Subject-independent emotion recognition of EEG signals based on dynamic empirical convolutional neural network," *IEEE/ACM Trans. Comput. Biol. Bioinf.*, vol. 18, no. 5, pp. 1710–1721, Sep. 2021.

[13] Y. An, S. Hu, X. Duan, L. Zhao, C. Xie, and Y. Zhao, "Electroencephalogram emotion recognition based on 3D feature fusion and convolutional autoencoder," *Frontiers Comput. Neurosci.*, vol. 15, p. 83, Oct. 2021.

[14] Y. Fang, H. Yang, X. Zhang, H. Liu, and B. Tao, "Multi-feature input deep forest for EEG-based emotion recognition," *Frontiers Neurorobotics*, vol. 14, p. 106, Jan. 2021.

[15] X. Zheng, M. Zhang, T. Li, C. Ji, and B. Hu, "A novel consciousness emotion recognition method using ERP components and MMSE," *J. Neural Eng.*, vol. 18, no. 4, Apr. 2021, Art. no. 046001.

[16] D. Li, B. Chai, Z. Wang, H. Yang, and W. Du, "EEG emotion recognition based on 3-D feature representation and dilated fully convolutional networks," *IEEE Trans. Cogn. Develop. Syst.*, vol. 13, no. 4, pp. 885–897, Dec. 2021.

[17] D. Huang, S. Chen, C. Liu, L. Zheng, Z. Tian, and D. Jiang, "Differences first in asymmetric brain: A bi-hemisphere discrepancy convolutional neural network for EEG emotion recognition," *Neurocomputing*, vol. 448, pp. 140–151, Aug. 2021.

[18] J. Cai, R. Xiao, W. Cui, S. Zhang, and G. Liu, "Application of electroencephalography-based machine learning in emotion recognition: A review," *Frontiers Syst. Neurosci.*, vol. 15, p. 146, Nov. 2021.

[19] A. Al-Nafjan, M. Hosny, A. Al-Wabil, and Y. Al-Ohali, "Classification of human emotions from electroencephalogram (EEG) signal using deep neural network," *Int. J. Adv. Comput. Sci. Appl.*, vol. 8, no. 9, pp. 419–425, 2017.

[20] B. Pan and W. Zheng, "Emotion recognition based on EEG using generative adversarial nets and convolutional neural network," *Comput. Math. Methods Med.*, vol. 2021, pp. 1–11, Oct. 2021.

[21] S. Liu et al., "3DCANN: A spatio–temporal convolution attention neural network for EEG emotion recognition," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 11, pp. 5321–5331, Nov. 2022.

[22] M. Ramzan and S. Dawn, "Fused CNN-LSTM deep learning emotion recognition model using electroencephalography signals," *Int. J. Neurosci.*, pp. 1–11, Aug. 2021.

[23] W.-L. Zheng and B.-L. Lu, "Investigating critical frequency bands and channels for EEG-based emotion recognition with deep neural networks," *IEEE Trans. Auton. Mental Develop.*, vol. 7, no. 3, pp. 162–175, Sep. 2015.

[24] T. Song, W. Zheng, C. Lu, Y. Zong, X. Zhang, and Z. Cui, "MPED: A multi-modal physiological emotion database for discrete emotion recognition," *IEEE Access*, vol. 7, pp. 12177–12191, 2019.

[25] Q. Kang et al., "Emotion recognition from EEG signals of hearing-impaired people using stacking ensemble learning framework based on a novel brain network," *IEEE Sensors J.*, vol. 21, no. 20, pp. 23245–23255, Oct. 2021.

[26] Z. Tian, D. Li, Y. Song, Q. Gao, Q. Kang, and Y. Yang, "EEG-based emotion recognition of deaf subjects by integrated genetic firefly algorithm," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–11, 2021.

[27] S. Koelstra et al., "DEAP: A database for emotion analysis; using physiological signals," *IEEE Trans. Affect. Comput.*, vol. 3, no. 1, pp. 18–31, Jan. 2012.

[28] W.-L. Zheng, W. Liu, Y. Lu, B.-L. Lu, and A. Cichocki, "EmotionMeter: A multimodal framework for recognizing human emotions," *IEEE Trans. Cybern.*, vol. 49, no. 3, pp. 1110–1122, Mar. 2019.

[29] J. A. Russell, "A circumplex model of affect," *J. Pers. Social Psychol.*, vol. 39, no. 6, pp. 1161–1178, Dec. 1980.

[30] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2117–2125.

[31] Y. Jiang, Z. Tan, J. Wang, X. Sun, M. Lin, and H. Li, "GiraffeDet: A heavy-neck paradigm for object detection," 2022, *arXiv:2202.04256*.

[32] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.

[33] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10781–10790.

[34] R. Jenke, A. Peer, and M. Buss, "Feature extraction and selection for emotion recognition from EEG," *IEEE Trans. Affect. Comput.*, vol. 5, no. 3, pp. 327–339, Jul./Sep. 2014.

[35] R.-N. Duan, J.-Y. Zhu, and B.-L. Lu, "Differential entropy feature for EEG-based emotion classification," in *Proc. 6th Int. IEEE/EMBS Conf. Neural Eng. (NER)*, San Diego, CA, USA, Nov. 2013, pp. 81–84.

[36] S. T. Aung and Y. Wongsawat, "Prediction of epileptic seizures based on multivariate multiscale modified-distribution entropy," *PeerJ Comput. Sci.*, vol. 7, Oct. 2021, Art. no. e744.

[37] H. Yang, J. Han, and K. Min, "A multi-column CNN model for emotion recognition from EEG signals," *Sensors*, vol. 19, no. 21, p. 4736, Oct. 2019.

[38] Y. Yin, X. Zheng, B. Hu, Y. Zhang, and X. Cui, "EEG emotion recognition using fusion model of graph convolutional neural networks and LSTM," *Appl. Soft Comput.*, vol. 100, Mar. 2021, Art. no. 106954.

[39] L. Feng, C. Cheng, M. Zhao, H. Deng, and Y. Zhang, "EEG-based emotion recognition using spatial–temporal graph convolutional LSTM with attention mechanism," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 11, pp. 5406–5417, Nov. 2022.

[40] N. Pusarla, A. Singh, and S. Tripathi, "Normal inverse Gaussian features for EEG-based automatic emotion recognition," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–11, 2022.

[41] X. Du et al., "An efficient LSTM network for emotion recognition from multichannel EEG signals," *IEEE Trans. Affect. Comput.*, vol. 13, no. 3, pp. 1528–1540, Jul. 2022.

[42] L. Chen, Q. She, M. Meng, Q. Zhang, and J. Zhang, "Similarity constraint style transfer mapping for emotion recognition," *Biomed. Signal Process. Control*, vol. 80, Feb. 2023, Art. no. 104314.

[43] F. Cui, R. Wang, W. Ding, Y. Chen, and L. Huang, "A novel DE-CNN-BiLSTM multi-fusion model for EEG emotion recognition," *Mathematics*, vol. 10, no. 4, p. 582, Feb. 2022.

[44] Y. Peng, F. Jin, W. Kong, F. Nie, B.-L. Lu, and A. Cichocki, "OGSSL: A semi-supervised classification model coupled with optimal graph learning for EEG emotion recognition," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 30, pp. 1288–1297, 2022.

[45] Z. Li et al., "Dynamic domain adaptation for class-aware cross-subject and cross-session EEG emotion recognition," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 12, pp. 5964–5973, Dec. 2022.

[46] Y. Yang, Q. Gao, Y. Song, X. Song, Z. Mao, and J. Liu, "Investigating of deaf emotion cognition pattern by EEG and facial expression combination," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 2, pp. 589–599, Feb. 2022.

Fazheng Hou received the B.S. degree in automation from the Tianjin University of Technology, Tianjin, China, in 2020, where he is currently pursuing the M.S. degree with the School of Electrical Engineering and Automation.

His current research interests include affective computing, signal processing, and deep learning.

Junjie Liu received the Ph.D. degree in control science and engineering from Nankai University, Tianjin, China, in 2020.

He is currently a Lecturer with the School of Electrical Engineering and Automation, Tianjin University of Technology, Tianjin. His current research interests include disturbance compensation-based control predictive control reinforcement learning and artificial intelligence.

Zhongli Bai received the B.S. degree in automation from the Tianjin University of Technology, Tianjin, China, in 2021, where he is currently pursuing the M.S. degree with the School of Electrical Engineering and Automation.

His current research interests include affective computing, signal processing, and machine learning.

Zhiyi Yang received the B.S. degree in electrical engineering and automation from Shenyang Urban Construction University, Shenyang, China, in 2020. He is currently pursuing the M.S. degree with the School of Electrical Engineering and Automation, Tianjin University of Technology, Tianjin, China.

His current research interests include affective computing and brain–computer interface.

Jiayin Liu received the B.S. degree in automation from Nanyang Normal University, Nanyang, China, in 2020. He is currently pursuing the M.S. degree with the School of Electrical Engineering and automation, Tianjin University of Technology, Tianjin, China.

His current research interests include multimodal affective computing and brain–computer interface.

Qiang Gao received the M.S. degree in control theory and control engineering from Tianjin University, Tianjin, China, in 1996.

He has been a Full Professor with the School of Electrical Engineering and Automation, Tianjin University of Technology, Tianjin, since 1996. His current research interests include intelligence controlling, brain–computer interface, and fault diagnosis.

Yu Song (Member, IEEE) received the Ph.D. degree in intelligent mechanical systems engineering from Kagawa University, Takamatsu, Japan, in 2018.

He is currently an Associate Professor with the School of Electrical Engineering and Automation, Tianjin University of Technology, Tianjin, China. His current research interests include haptic control, human–robot interaction, teleoperation systems, brain–computer interface, and artificial intelligence.